

Question M

Longest Consecutive Sequence

Solved

Given an array of integers `nums`, return *the length* of the longest consecutive sequence of elements that can be formed.

A *consecutive sequence* is a sequence of elements in which each element is exactly 1 greater than the previous element. The elements do *not* have to be consecutive in the original array.

You must write an algorithm that runs in $O(n)$ time.

Example 1:

Input: `nums = [2,20,4,10,3,4,5]`

Output: 4

Explanation: The longest consecutive sequence is [2, 3, 4, 5].

Example 2:

Input: `nums = [0,3,2,5,4,6,1,1]`

Output: 7

Constraints:

- $0 \leq \text{nums.length} \leq 1000$
- $-10^9 \leq \text{nums}[i] \leq 10^9$

```
class Solution:
```

```
    def longestConsecutive(self, nums: List[int]) -> int:
```

```
        if not nums:
```

```
            return 0
```

```
        r = []
```

```
        # print('start of the ',temp, t[0])
```

```
        for e, i in enumerate(sorted(set(nums))):
```

```
            if e == 0:
```

```
                temp = [i]
```

```
            if i + 1 in nums :
```

```
                print('i before adding',i, 'i after adding', i+1)
```

```
                print('adding : ', i+1, "to temp.")
```

```
                temp.append(i+1)
```

```
                print(temp)
```

```
                print('r, ',r)
```

```
            else:
```

```
                print('entred else,' , r)
```

```
                r.append(len(temp))
```

```
                temp = [i+1]
```

```
        r.append(len(temp))
```

```
        return max(r)
```

